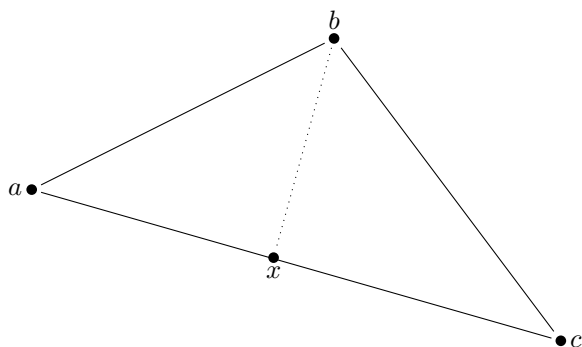


CSCI 340, Winter 2023, Math Homework # 2

Name _____

Typeset this homework using L^AT_EX.

1. In the picture below, a , b , and c are arbitrary points, and the dotted line from b to x is perpendicular to the line from a to c . Give formulas to find each of the distances from a , b , and c to x as a function of the points a , b , and c . Use point subtraction and dot products. Each formula should stand on its own and not depend on the other formulas.

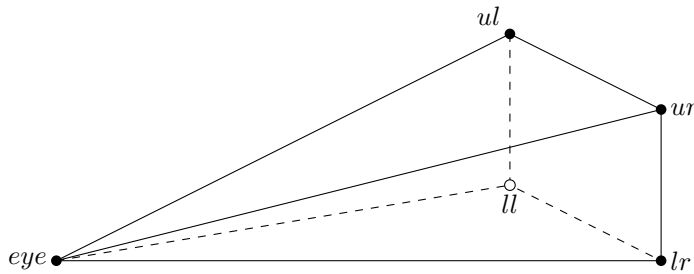


Distance a to x :

Distance b to x :

Distance c to x :

2. Suppose we specify a camera by five points: the eye position and the four corners of the image plane (upper left, upper right, lower left, lower right), as in the figure below (the left side of the view plane is deeper into the picture than the right side).



Given a position in the image plane defined by $percentX$ and $percentY$ (in percent coordinates), and with the origin of the image plane understood as the upper left corner, write an expression giving the vector for a ray from the eye to that point in the view plane. You do not need to normalize the vector.

3. Given a camera specified as in the lecture notes, with an eye position, normalized right, up, and forward vectors, and scalars depth, width and height, $\langle p, r, u, f, d, w, h \rangle$, write expressions for each of the four corner points in the camera representation from the previous problem and the center/focus.

(a) $focus/center =$

(b) $ul =$

(c) $ur =$

(d) $ll =$

(e) $lr =$

4. For each of the following implicitly defined quadric surfaces, find formulas for the coefficients for the quadratic equation, $at^2 + bt + c = 0$ to determine the value of t where a ray defined by $p + tv$ intersects the surface.

(a) Elliptic paraboloid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - z = 0$$

$$a =$$

$$b =$$

$$c =$$

(b) Hyperbolic paraboloid

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} - z = 0$$

$$a =$$

$$b =$$

$$c =$$

(c) Elliptic hyperboloid of one sheet

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

$$a =$$

$$b =$$

$$c =$$

(d) Elliptic hyperboloid of two sheets

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1$$

$$a =$$

$$b =$$

$$c =$$

5. Find a formula for a vector normal to each of the following surfaces, given a point (x, y, z) on the surface.

(a) Elliptic paraboloid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - z = 0$$

(b) Hyperbolic paraboloid

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} - z = 0$$

(c) Elliptic hyperboloid of one sheet

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

(d) Elliptic hyperboloid of two sheets

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1$$